



## Koneru Lakshmaiah Education Foundation

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# STUDENT CASE STUDY EXECUTION REPORT

|                |                          |
|----------------|--------------------------|
| Department     | FED                      |
| Subject        | DSA-2                    |
| Academic Year  | I-III Semester (2025-26) |
| Faculty Name   | G.Mamatha                |
| Student Name   | Penmatsa Gowtham Varma   |
| Roll Number    | 2520090202               |
| Branch         | CSIT                     |
| Course Outcome | CO1                      |

**Case Study Topic:** Recursive DNS Resolver – TTL Driven Cache Expiry using Min Heap and HashMap

### Work Done Summary:

In this case study, a DNS cache management system was implemented using a Min Heap and HashMap. The system stores domain records along with their expiration times (TTL) and efficiently removes expired entries without scanning the entire cache.

### Implementation Details:

- Created a CacheEntry class to store domain name, expiration time, and DNS record.
- Implemented a Min Heap (Priority Queue) to maintain entries based on expiration time.
- Used a HashMap for fast domain lookup operations.
- Inserted DNS records into both Min Heap and HashMap.
- Implemented cache eviction using heap root checking.
- Removed expired records from both data structures.
- Handled TTL renewal using updated cache entries.
- Displayed evicted domains after expiration processing.

## Result: Screenshots/Output

### Output

DNS CACHE EXPIRY SYSTEM

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Current Time : 1700000030

Expired Domains:

kl.ac.in

hdfc.in

Total Expired Domains = 2

=== Code Execution Successful ===

### Observations:

DNS records were successfully inserted into the cache.

- Domain lookup was performed efficiently using HashMap.
- Expired entries were removed correctly using Min Heap operations.
- Cache eviction was faster than periodic full-cache scanning.
- The Min Heap always provided the next entry to expire.
- The implementation reduced cache maintenance overhead.

GitHub Link: <https://github.com/gowthampenmatsa07-prog/D-S-A.git>

| Student Signature | Faculty Signature |
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